

AZIZ ASHFAK

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RESEARCH INTERESTS

Trustworthy and evaluable NLP: retrieval-augmented generation, automated causal-knowledge extraction, benchmark and evaluation design for large language models, low-resource (Bangla) NLP, and statistical methodology for reproducible, decision-relevant AI.

EDUCATION

Noakhali Science and Technology University — Noakhali, Bangladesh
Bachelor of Science (Honours) in Statistics | CGPA: **3.30 / 4.00**

Apr 2022 – Aug 2026

RESEARCH EXPERIENCE

1. Gulf Council Research Project — Research Collaborator (AI / ML) **Nov 2025 – Present**

Evidence-grounded causal knowledge graphs from public fisheries discourse · [project repository](#)

- **Corpus pipeline** — crawls, screens and structures public discourse on Gulf of Mexico fisheries into 3,000+ documents from 122 sites, indexed as 498,770 embedded passages with full provenance (source, URL, date) for every document.
- **Retrieval system** — RAG for evidence-grounded question answering (BGE embeddings, exact inner-product FAISS search, LLM generation via LangChain), tying every answer to retrievable source passages rather than model memory.
- **Causal extraction** — signed causal triples with verbatim quotations — 112,378 raw claims normalised through a 45-concept, 309-pattern vocabulary into a reference Fuzzy Cognitive Map of 909 evidence-thresholded edges, stored in PostgreSQL for centrality, feedback-loop and scenario-simulation analysis.
- **Methodological audit** — twelve experiments with pass criteria fixed before results were inspected: 96.3% of claims recoverable in their cited document against a 0.2% shuffled-document control, concept ranking stable across 23 configuration perturbations (worst-case rank agreement 0.971), and 98.2% recall of a precision-first silver standard at 83.3% sign accuracy.
- **Limits reported, not hidden** — the concept dictionary discards 86.1% of raw claims (permutation-tested as narrowness, not bias), and the headline retrieval score shifts 11.2 points in precision between two independent LLM judges ($\kappa = 0.456$) — a judge-sensitivity analysis now reported as standard practice.
- **Benchmark** — frozen, fingerprinted dataset (200 corpus-verified questions over 36 concepts, 170 stratified passages, 3,578 perturbed variants, 1,243 reference edges); five OpenAI models scored against the RAG system and a closed-book control, putting retrieval's contribution to causal accuracy at **+0.136 F1 (95% CI [0.102, 0.174])**, with no system reproducing half its own causal structure when asked the same question twice.
- **Reproducibility** — seeded generation, SHA-256 fingerprints binding each result to its dataset, and notebooks that regenerate every figure and statistic without further model calls.

Technologies: Python, PostgreSQL, LangChain, OpenAI API, Hugging Face Transformers, sentence-transformers (BGE), FAISS, PyTorch, PEFT, RAG, Fuzzy Cognitive Maps, NetworkX, Jupyter.

2. Human–Wildlife Conflict Corpus & Fuzzy Cognitive Map Project — Research Data Collaborator **Jun 2026 – Present**

Comparative mental models of human–wildlife conflict across 19 South and Southeast Asian countries

- **Corpus construction** — dual-pipeline Python ETL that collects, filters, classifies and archives conflict literature and public discourse across 19 countries into a structured corpus of 5,300+ documents.
- **Source engineering** — 10 API connectors (OpenAlex, Crossref, Europe PMC, DOAJ, OpenAIRE, GDELT, Reddit, RSS news, podcasts) with rate-limit handling, budget-aware failover and resumable runs, so a degraded source never stalls collection.
- **Quality control** — multi-stage gate — weighted relevance scoring, a 19-country/150-landscape gazetteer and four-way deduplication — logging every rejection by reason, plus a two-axis taxonomy separating document type (16 categories) from research theme.
- **Causal extraction** — LLM pipeline (OpenAI API, PostgreSQL on Docker) converting thousands of stakeholder documents into Fuzzy Cognitive Maps: 60,000+ concept nodes and signed causal claims, each traceable to its exact source document and URL.
- **Network analysis** — ten stakeholder- and region-partitioned maps compared in NetworkX via centrality metrics, Jaccard similarity, Mann–Whitney U tests, bootstrap resampling and FDR correction, locating where mental models converge and diverge.
- **Output** — APA-7 manuscript and conference abstract, with publication-ready figures and tables generated from a reproducible Jupyter pipeline backed by a 178-assertion validation suite.

Technologies: Python (pandas, NumPy, SciPy, NetworkX, BeautifulSoup, SQLAlchemy), PostgreSQL, Docker, OpenAI API, Jupyter, Git.

3. Coastal Evidence Synthesis Project — Research Collaborator **Aug 2026 – Present**

Where coastal planning discourse diverges from the science it cites

- **Document pipeline** — extracted and repaired 103 planning, health and environmental PDFs (12,257 pages; 3.3M words), coding 8,940 segments against a 44-code, 9-theme codebook at ~94% audited precision.
- **Thematic structure** — segment-level co-occurrence (lift) showing Public Health co-occurs below chance with all five other themes (0.26–0.60) — a structural environment–health disconnect in the planning corpus.

- **Evidence base** — 744 cited references recovered, 607 Crossref-verified, open access resolved via Unpaywall/OpenAlex and 159 full texts retrieved; coding them with the same codebook showed all nine themes discussed more than evidenced (widest gap: water quality, 72.8% of documents vs 2.9% of cited science).
- **Delivery** — self-contained interactive dashboard — 26 figures and tables, 12 published datasets and full provenance for every number — verified by automated cross-browser and accessibility tests.

Technologies: Python (pandas, PyMuPDF, SQLite), Crossref/Unpaywall/OpenAlex APIs, JavaScript, Playwright, Jupyter.

4. Undergraduate Research Project — BanglaStatMCQ-500: A Bangla Reasoning Benchmark May 2026 – Aug 2026

Supervisor: Md. Rasel Hossain, Assistant Professor, Department of Statistics

- **Dataset** — curated and validated 500 multiple-choice items spanning probability, hypothesis testing, regression, statistical inference and experimental design, with quality assurance at every stage of construction.
- **Evaluation** — designed the evaluation framework and ran comparative experiments across multiple open-source LLMs, analysing reasoning ability, robustness and error patterns in domain-specific Bangla NLP.

Technologies: Python, PyTorch, Hugging Face Transformers, NLP, LLM evaluation, pandas, NumPy.

PUBLICATIONS & MANUSCRIPTS

Accepted / Peer-Reviewed

- Optimizing U-Net Performance: Evaluation of State-of-the-Art Encoders and Optimizers on Human Segmentation Dataset. 2026 *IEEE International Conference on Systems, Man, and Cybernetics (SMC 2026)*, Bellevue, WA, Oct. 2026. **Accepted (Regular Paper); to appear in IEEE Xplore.** [co-author]

Under Review

- Robustness of CNN Transfer Learning versus Vision Transformers for Aerial Scene Classification. *Submitted to ICCIT 2026.* **[first author]**
- Self-Supervised versus ImageNet Pretraining for Bangladeshi Traffic Vehicle Recognition under Limited Labels. *Submitted to IEEE BDCloud 2026.* **[first author]**
- Transfer Learning for Chest X-Ray Lung Segmentation: A Comparative Study of CNN Backbones. *Under review, Scientific Reports.* [co-author]
- Skilled Birth Attendance and Height-for-Age Among Children Aged 0–23 Months in Bangladesh: A Survey-Weighted Targeted Maximum Likelihood Analysis of the 2025 Multiple Indicator Cluster Survey. *Under review, BMC Public Health.* [co-author]
- Functional Difficulty and ICT Skill Attainment Among Women in Bangladesh: Evidence from the 2025 Multiple Indicator Cluster Survey. *Under review, BMC Public Health.* [co-author]
- Same Score, Different Deprivation: A Survey-Weighted Latent Class Analysis of Multidimensional Deprivation Among Children Aged 24–59 Months in Bangladesh. *Under review, BMC Public Health.* [co-author]

In Preparation

- Can a Retrieval-Augmented Fuzzy Cognitive Map Be Trusted? A Twelve-Experiment Audit of Gulf of Mexico Fisheries Discourse. [co-author]
- Retrieval Grounds Causal Maps: A Benchmark for Fuzzy Cognitive Maps Extracted from Language-Model Answers about Gulf of Mexico Fisheries. [co-author]
- BanglaStatMCQ-500: A Statistical Reasoning Benchmark in Bangla for Open-Source Large Language Models. **[first author; undergraduate research project]**
- Comparative Mental Models of Human–Wildlife Conflict across South and Southeast Asia: A Fuzzy Cognitive Map Analysis. [co-author; APA-7 manuscript + conference abstract]

PROFESSIONAL EXPERIENCE

AI Engineer — ROYALX LLC, Dhaka, Bangladesh (remote, full-time) Mar 2025 – Nov 2025

- Developed and deployed LLM applications with OpenAI, Groq, Hugging Face, LangChain and LlamaIndex, and fine-tuned transformer models for question answering, classification and document understanding.
- Built retrieval-augmented generation pipelines (including multimodal inputs) and LangChain agents for multi-step reasoning, shipped as production FastAPI, Flask and Streamlit services.
- Delivered client-facing AI systems with remote teams, tuning latency, robustness and usability, and integrating LLM workflows into existing enterprise applications.

AI Tutor — AI Devlop, Dhaka, Bangladesh (remote, part-time) Apr 2025 – Aug 2025

- Taught 100+ students exploratory data analysis, statistical analysis, machine learning and deep learning, mentoring them through end-to-end ML pipelines from preprocessing to deployment.

Freelance Data Analyst — Hestia Traffic App Project Dec 2024 – Feb 2025

- Analysed large-scale traffic and road-accident datasets to identify the factors driving crash severity, using cleaning, preprocessing, exploratory analysis and statistical testing of influential variables (pandas, NumPy, Matplotlib).
- Supported the analytical evaluation of a data-driven platform for monitoring traffic conditions, translating findings into actionable safety recommendations.

SELECTED PROJECTS

- Multimodal Research Paper Analyst** ([code](#), [demo](#)) — Multimodal RAG system that ingests research papers and answers questions over both their text and their figures, so charts and diagrams are interpreted rather than skipped. Modular retrieval and generation pipelines behind a Streamlit interface, with custom queries and structured report export to Word, PDF, CSV and Markdown. Built because text-only retrieval silently drops the figures that carry a paper's actual results.
Tech: Python, Groq API, Llama, LangChain, multimodal RAG, vector retrieval, Streamlit.
- Brain Tumor Detection** ([code](#), [live app](#), [demo](#)) — 98.52% accurate MRI brain-tumour classifier built by comparing three transfer-learning backbones (LeNet, ResNet152V2, DenseNet121) under matched normalisation and augmentation, then exporting the winner to ONNX for low-latency inference behind a deployed web service with GitHub Actions CI/CD. Backbones were compared under identical augmentation and validation splits, so the reported gap reflects architecture rather than training luck.
Tech: Python, TensorFlow/Keras, transfer learning, ONNX Runtime, Flask, GitHub Actions, Render.
- Indian Flight Price Prediction** ([code](#), [live app](#)) — 99.69% accurate fare-prediction model (BaggingRegressor) trained on a 300,261-record airline dataset, with a reproducible preprocessing pipeline (MinMaxScaler, OneHotEncoder, SimpleImputer) and automated deployment through GitHub Actions. Built to compare ensemble regressors under identical preprocessing so the accuracy gain is attributable to the model rather than the feature pipeline.
Tech: Python, scikit-learn, pandas, NumPy, Flask, GitHub Actions CI/CD, Render.
- Thyroid Disease Prediction** ([code](#), [live app](#)) — 98.7% accurate thyroid-disease classifier on the UCI dataset using a hyperparameter-tuned RandomForest, with missing-value imputation and categorical encoding handled in a single reproducible pipeline and served for real-time prediction. Clinical tabular data with substantial missingness, so imputation strategy drives performance as much as model choice.
Tech: Python, scikit-learn, pandas, Flask, GitHub Actions CI/CD, Render.
- Face Mask Detection** ([code](#)) — Real-time mask-compliance detector trained with YOLO transfer learning on an 853-image annotated dataset, exported to ONNX for optimised inference and served through a Flask application for live monitoring in workplaces and public transport. Object detection rather than classification, so the model localises each face and scales to multiple people per frame.
Tech: Python, YOLO, OpenCV, transfer learning, ONNX Runtime, Flask.
- QnA RAG Bot** ([code](#), [demo](#)) — Retrieval-augmented question answering over user-uploaded PDFs, returning context-grounded answers tied to the retrieved passages rather than model memory. FAISS vector search over chunked documents, figure and graph analysis, and multi-format export (Markdown, CSV, DOCX, PDF), deployed as a modular service. The retrieval layer is swappable, so chunking and embedding choices can be compared without touching the generation code.
Tech: Python, FastAPI, LangChain, Hugging Face Transformers, FAISS, RAG, Render deployment.

TECHNICAL SKILLS

Languages Python, SQL, R, C (basic), Java (basic)

CS Foundations Data structures and algorithms, object-oriented programming, model evaluation, REST API design, version control and Git workflows, CI/CD

ML & DL PyTorch, TensorFlow, Keras, scikit-learn, OpenCV, CNNs, Transformers, Vision Transformers, transfer learning, self-supervised pretraining, fine-tuning, PEFT/LoRA

LLMs & RAG OpenAI API, Hugging Face Transformers, LangChain, LangGraph, LlamaIndex, Groq API, CrewAI, multimodal and agentic RAG, knowledge-graph RAG, prompt engineering, AI agents

Statistics Hypothesis testing, regression, statistical inference, experimental design, bootstrap resampling, permutation tests, FDR correction, inter-rater agreement, network and centrality analysis, Fuzzy Cognitive Maps

Databases & Retrieval PostgreSQL, SQLite, schema design for provenance-tracked corpora, FAISS, ChromaDB, Pinecone, exact and approximate nearest-neighbour search, SQLAlchemy

Data Engineering pandas, NumPy, SciPy, NetworkX, PyMuPDF, BeautifulSoup, ETL pipeline design, API integration with rate-limit handling and failover, deduplication and quality gating, large-scale text corpus construction

Visualisation Matplotlib, publication-ready figure and table generation, interactive HTML/JavaScript dashboards, network and graph visualisation, automated cross-browser and accessibility testing with Playwright

Research Methods Benchmark and evaluation design, pre-registered pass criteria, ablation and perturbation testing, LLM-as-judge evaluation with judge-sensitivity analysis, silver-standard construction, provenance tracking, reproducible pipelines

Software Engineering REST API design, modular pipeline architecture, containerisation with Docker, automated testing and validation suites, resumable and fault-tolerant batch jobs, version control and CI/CD with GitHub Actions, ONNX model export and optimised inference

ACHIEVEMENTS & SERVICE

- **General Secretary** — NSTU Data Science Club
- **Research Instructor** — NSTU Research Society
- **Treasurer** — NSTU Blood Brigade Society

Feb 2026 – Aug 2026

Jan 2024 – Aug 2025

Jan 2023 – Nov 2023